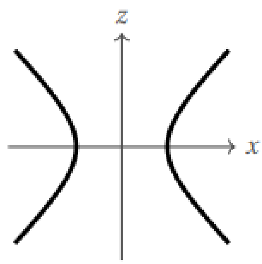
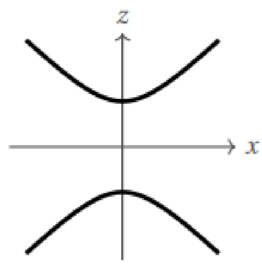
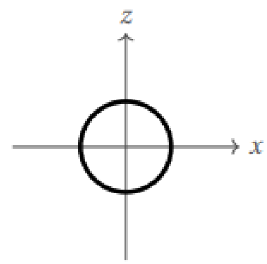
MTH 202 Chapter 1: Three-Dimensional Space & Vectors

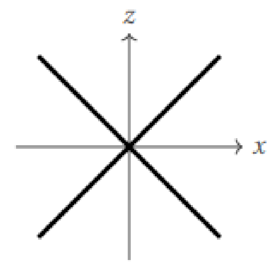
1.7 Quadric Surfaces Worksheet

1. Consider the quadric surface Q defined by the equation $x^2 + y^2 - z^2 = 1$. Check the box below the picture of the curve formed by intersecting Q with the xz-plane.

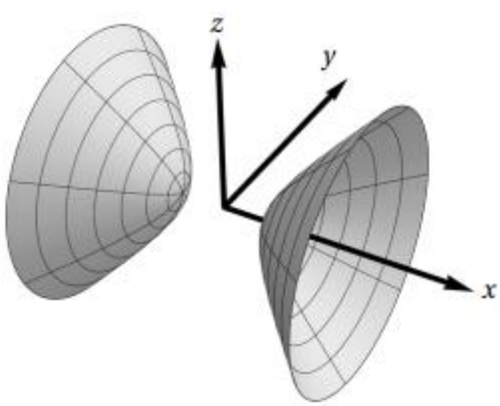








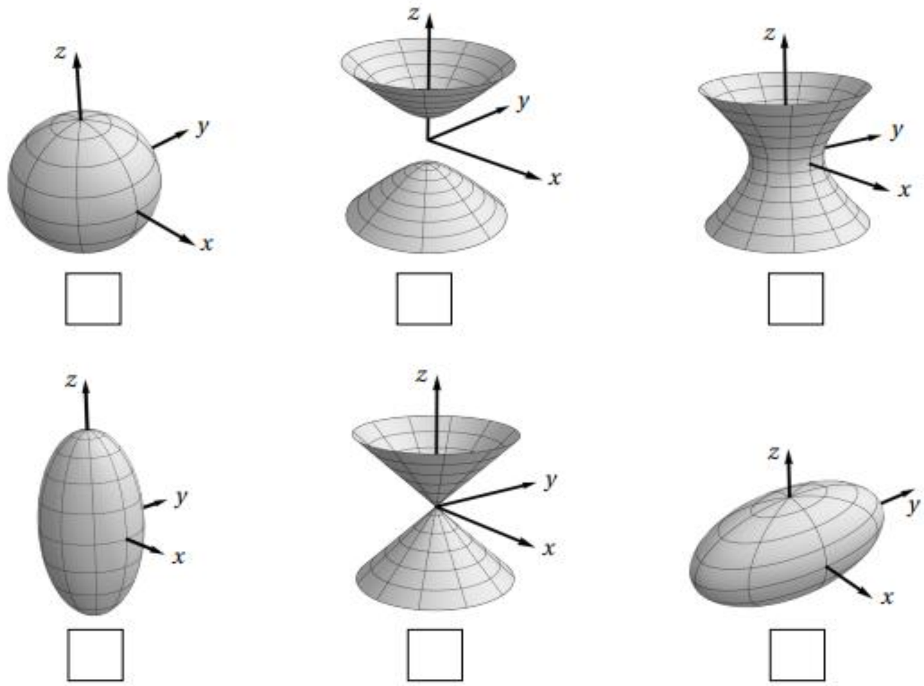
2. Consider the hyperboloid H of two sheets shown below. Circle the equation of H.



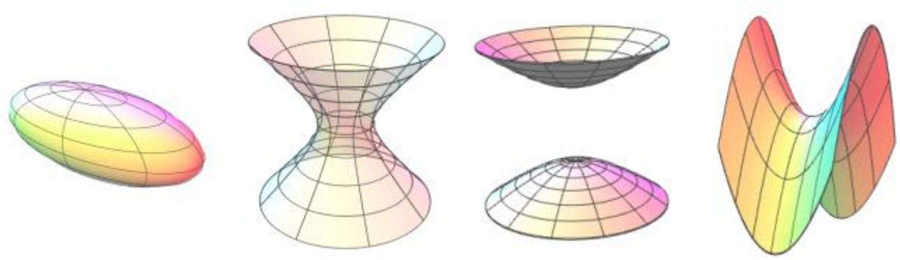
$y^2 + z^2 - x^2 = 1$	$x = y^2 + z^2 + 1$
$x^2 + y^2 + z^2 = -1$	$x^2 - y^2 - z^2 = 1$

3. For each equation below, write the corresponding letter in the box next to the picture of the surface it describes.

(A) $x^2 + y^2 - z^2 + 1 = 0$ (B) $4x^2 + y^2 + 4z^2 - 1 = 0$

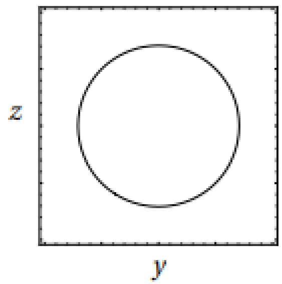


4. Let $f(x, y, z) = ax^2 + by^2 + cz^2$ for some real numbers a, b, and c. Which of the following *could not* be a level set of f?

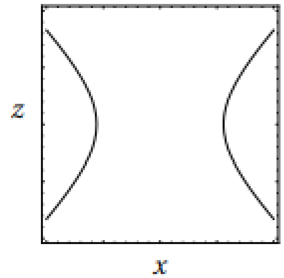


A. Ellipsoid **B.** Hyperboloid of 1 sheet **C.** Hyperboloid of 2 sheets **D.** Hyperbolic paraboloid

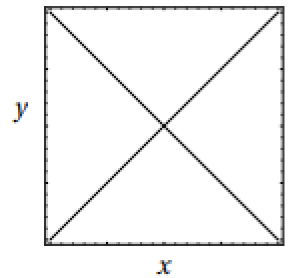
5. Suppose S is a quadric surface whose intersections with the following three planes are:



Plane $\{x = 1\}$

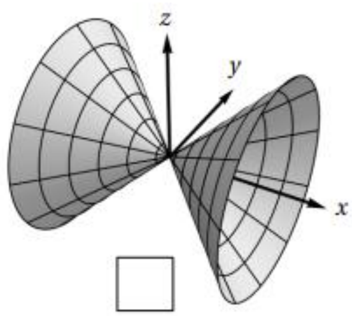


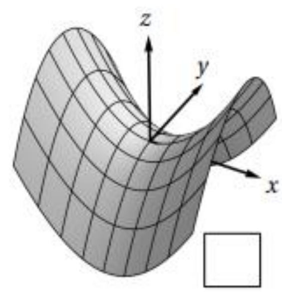
Plane $\{y = -1.5\}$

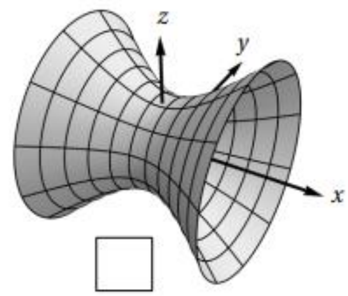


Plane $\{z = 1\}$

a) Mark the box next to the picture of the portion of S where $-2 \le x \le 2$, $-2 \le y \le 2$, and $-2 \le z \le 2$.







b) Circle the equation that S satisfies:

$x^2 + y^2 - z = 0$	$y^2 + z^2 - x^2 = 1$	$y^2 + z^2 = x^2$

6. Determine quadric surface is represented by each of the following equations.

a) $x^2 + y + 2z^2 = 1$

b) $10x^2 + 4y^2 - z^2 = 0$

c) $\frac{1}{2}x + y^2 - z^2 = 2$